Assessing musculoskeletal system

Weber: Chapter 24

NURS3120: Health Assessment and Promotion

Objectives

1. Describe the structure and function of the musculoskeletal system
2. Differentiate between normal and abnormal findings of the musculoskeletal system
3. Analyze interview and physical assessment data related to the musculoskeletal system to formulate valid clinical judgments

Structure and function: Bones



Purpose: Provide structure and movement for body parts

Bones

Structure & protection	Serve as levers	Store calcium	Produce blood cells

2 types

Compact	Spongy

Bones provide us structure and protection, serve as levers, store calcium, and produce blood cells. We have a total of 206 bones that make up our skeleton.

Compact bone: Hard/dense, makes up the shaft and outer layers.
Spongy bone: Contains numerous spaces, makes up the ends and centers of the bones.

Structure and function: Muscles

Skeletal muscles

- **Conscious control**
- **Made of long muscle fibers arranged together in bundles & joined by connective tissue**
- **Attach to bones by tendons**
- **Assist with posture, produce body heat, allow body to move**

3 types of muscles: Skeletal, smooth, and cardiac. This chapter is focusing on skeletal.

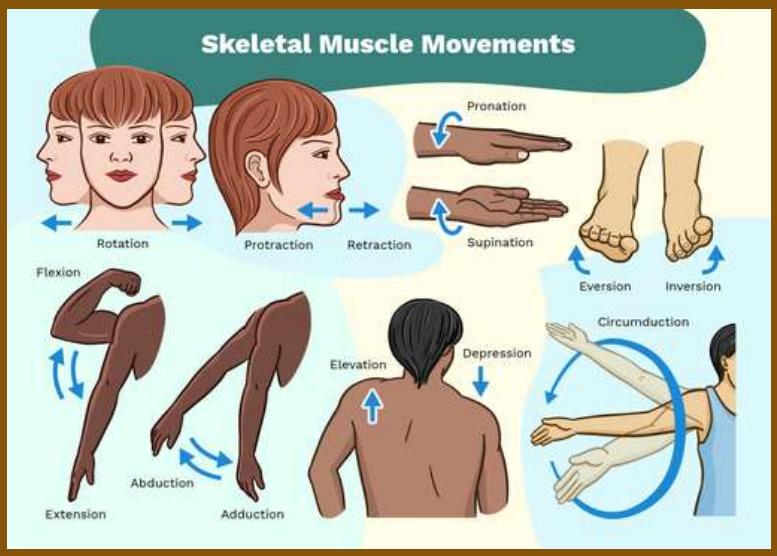
Tendons: Strong fibrous cords



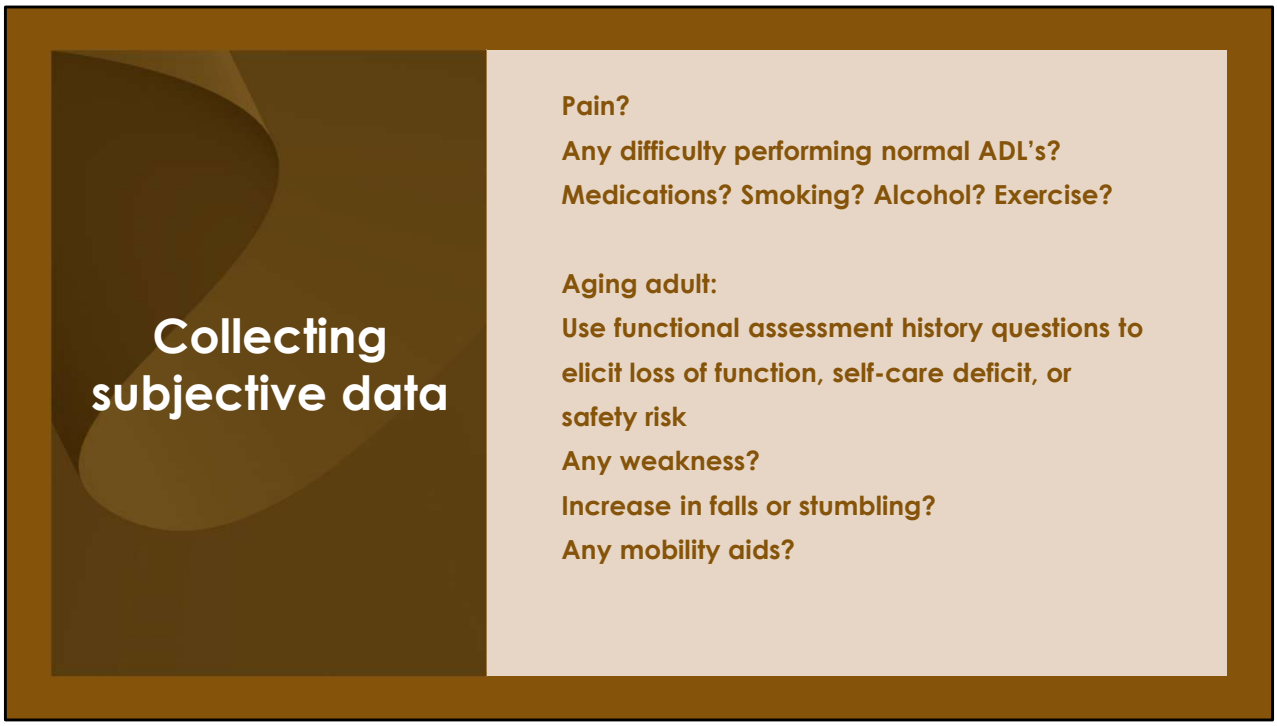
Structure and function: Joints

- Joint: Where 2 or more bones meet
- Classified as:
  - Fibrous
  - Cartilaginous
  - Synovial
    - Joined by ligaments

Fibrous joints (e.g., sutures between skull bones) are joined by fibrous connective tissue and are immovable.
Cartilaginous joints (e.g., joints between vertebrae) are joined by cartilage.
Synovial joints (e.g., shoulders, wrists, hips, knees, ankles; contain a space between the bones that is filled with synovial fluid, a lubricant that promotes a sliding movement of the ends of the bones. Bones in synovial joints are joined by ligaments, which are strong, dense bands of fibrous connective tissue



Abduction: Moving away from the midline of the body
Adduction: Moving towards midline of the body
Circumduction: Circular motion
Inversion: Moving inward
Eversion: Moving outward
Extension: Straightening of the extremity at the joint and increasing the angle of the joint
Hyperextension: joint bends greater than 180 degrees
Flexion: bending the extremity at the joint and decreasing the angle of the joint
Pronation: turning or facing downward
Supination: Turning or facing upward
Protraction: Moving forward
Retraction: moving backward
Rotation: turning of a bone on its own long axis
Internal rotation: turning of a bone toward the center of the body
External rotation: turning of a bone away from the center of the body

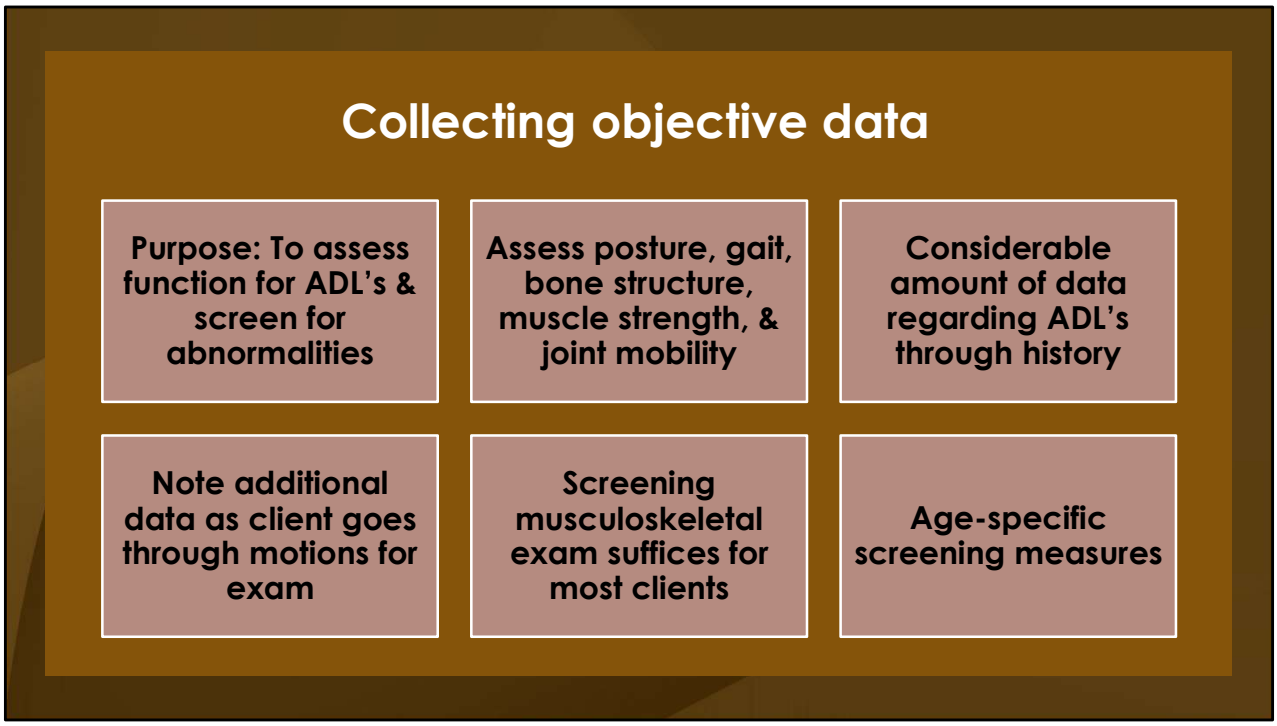


Assessment of the musculoskeletal system helps to evaluate the client's level of functioning with activities of daily living (ADLs). This system affects the entire body, from head to toe, and greatly influences what physical activities a client can and cannot do. Only the client can give you data regarding pain, stiffness, and levels of movement and how ADLs are affected. In addition, information regarding the client's nutrition, activities, and exercise is a significant part of the musculoskeletal assessment. Pain or stiffness is often a chief concern with musculoskeletal problems; therefore, a pain assessment may also be needed.

Bone pain is often dull, deep, and throbbing. Joint or muscle pain is described as aching, but has been differentiated between mechanical and inflammatory type pains.
Sharp, knife-like pain occurs with most fractures and increases with motion of the affected part.
Osteoarthritis pain usually begins in one set of joints and on one side of the body, with a feeling of pain deep in the joints. Stiffness is usually seen in the morning and with activity through out the day the stiffness will improve.
Do you exercise, do you drink alcohol or caffeine, do you smoke? Smoking/alcohol/ and caffeine increases the risk of osteoporosis. Regular exercise promotes flexibility,

muscle tone, and strength. Weight-bearing exercises promote bone density.
Any difficulty performing normal activities of daily living: sometimes people just need the use of assistive devices to perform ADL's and also ensure safety and independence.

Aging Adult: Bones lose their density with age, putting the older client at risk for bone fractures, especially of the wrists, hips and vertebrae. Older clients who have osteomalacia or osteoporosis are at an even greater risk for fractures. Joint stiffness conditions may be misdiagnosed as arthritis, especially in the older adult.



Physical assessment of the musculoskeletal system provides data regarding the client's posture, gait, bone structure, muscle strength, and joint mobility, as well as the client's ability to perform ADLs.

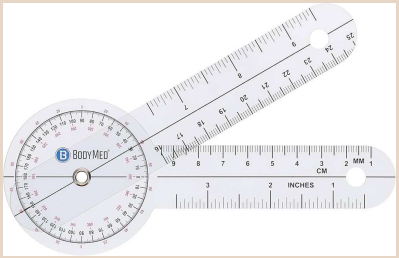
The physical assessment includes inspecting and palpating the joints, muscles, and bones; testing ROM; and assessing muscle strength.

You will already have a considerable amount of data from the client's history

Note additional ADL data as client goes through motions necessary for exam: how they sit in chair, get up from chair, takes off jacket, manipulates small objects like a pen, raises from supine position

Age-specific, screening for risk of falls

Collecting objective data



Inspect

- **Size & contour of joint**
- **Skin color & characteristics**

Palpate

- **Skin, muscles, bony articulations, & joint capsules**

Range of motion (ROM)

- **Active**
- **Passive**
- **Measure with goniometer if abnormal**

Muscle strength

When you are inspecting each joint and size; you are looking for any abnormalities. They are the same inspection at each joint: notice for any scars, color, swelling, or masses.
When palpating each joint you are looking for tenderness, swelling, heat or nodules
Active ROM the patient can perform; passive you perform

Muscle strength



- **Muscle strength graded on a 0 to 5 scale**
  - **0: No muscular contraction noted**
  - **1: Slight flicker of contraction**
  - **2: Passive ROM; gravity removed & assisted by examiner**
  - **3: Active movement against gravity**
  - **4: Active motion against some resistance**
  - **5: Active motion against full resistance**

0: Paralysis
1: Severe weakness
2: Poor ROM
3: Average weakness
4: Slight weakness
5: Normal

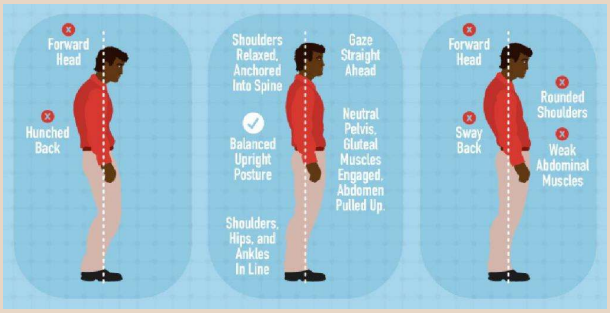
Range of motion

- Ask client to perform active ROM
- Be familiar with the type of each joint & its normal ROM
- For limitations, gently attempt passive ROM
- Joint motion normally causes no tenderness, pain, or crepitation
- Do not confuse crepitation with normal discrete "crack" heard as tendons or ligaments slip over bone during motion

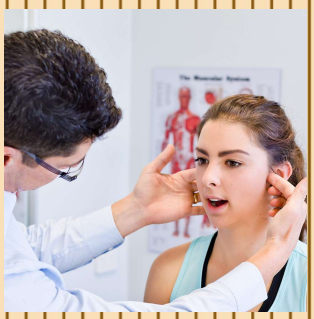
Crepitation:
Bone-on-bone rubbing due to cartilage wear, air bubbles (like knuckle cracking), or tendon/ligament issues.

Inspect: Posture & gait

- **Posture: Note alignment of head, trunk, pelvis, & extremities**
- **Gait: Note base of support, weight-bearing stability, foot position, stride, arm swing**
- **Assess risk for falls: Perform the "Nudge test"**



Inspection of Posture and Gait:
Observe posture which should be erect and comfortable for age. Abnormal: lordosis, scoliosis, or kyphosis.
Observe Gait: Can be done when the client walks into the room, should be smooth and equal on both sides. Abnormal is uneven weight bearing, client cannot stand on heels or toes, toes point inward, wide based gait.
The nudge test assess the risk of falling by standing behind the client and putting your arms around the client while you gently nudge the sternum. Normal: the client does not fall, abnormal: falls back easily can be seen with cervical spondylosis and Parkinson disease.



Inspect & palpate: Temporomandibular joint

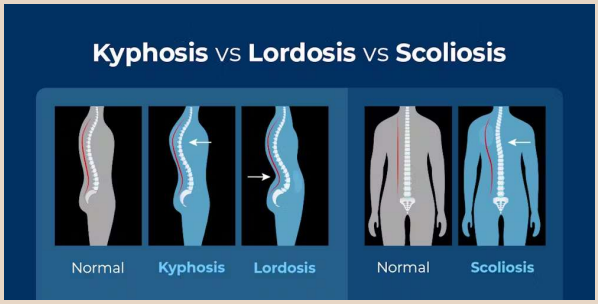
- Place tips of first two fingers in front of each ear; ask client to open & close mouth
- Note smooth motion of mandible
- Ask client to move jaw forward & laterally against resistance and to open mouth against resistance

Movement of the jaw: Open/close, protract/retract, lateral side-to-side

Snapping & clicking may be felt & heard in the normal client.

Arthritis may cause decrease ROM, swelling, tenderness, or crepitus
TMJ dysfunction may cause decrease ROM, a clicking, popping, or grating sound

Inspect, palpate, & ROM: Cervical, thoracic, & lumbar spine



Movements of the spine: Flexion/hyperextension, lateral bending, rotation

Observe the cervical, thoracic, & lumbar curves.
Cervical and lumbar spines are concave; thoracic spine is convex. Spine is straight (when observed from behind).

Kyphosis: Exaggerated thoracic curve. Common with aging.
Lordosis: Exaggerated lumbar curve. Seen in pregnancy or obesity.
Scoliosis: Lateral curvature of the spine.

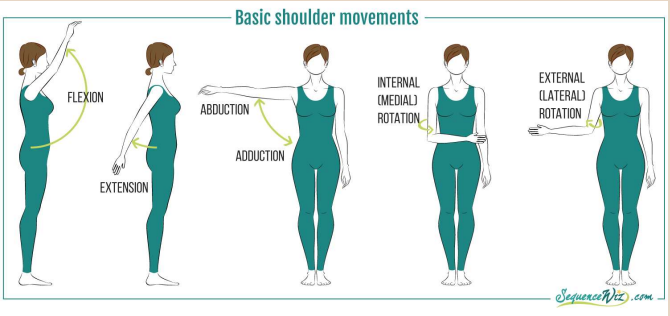
Palpation: Joints should be non-tender.

Cervical strain is the most common cause of neck pain. Impaired ROM and pain from anomalies of the soft tissue (muscles, ligaments, nerves) due to straining or injuring the neck. (Sleeping in the wrong position, carrying a heavy item, motor vehicle accident). You can also see a decrease in ROM of the neck along with fever, chills, and headache in meningitis.

Low back strain from injury to the soft tissues is a common cause of impaired ROM

and pain in the lumbar and thoracic regions. Other causes may include: osteoarthritis, ankylosing spondylitis, & congenital anomalies.

Inspect, palpate, & ROM: Shoulders



Movements of the shoulder: Flexion/extension, abduction/adduction, circumduction, internal & external rotation

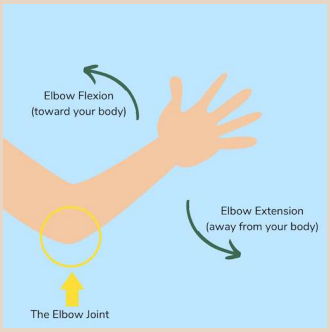
Articulation of the head of the humerus in the glenoid cavity of the scapula. Ball-and-socket joint.

Inspect & palpate shoulders and arms.
Shoulders are symmetrically round; no redness, swelling, structural abnormality, or heat. Muscles are fully developed. Clavicles and scapulae are even and symmetric. The client reports no tenderness.

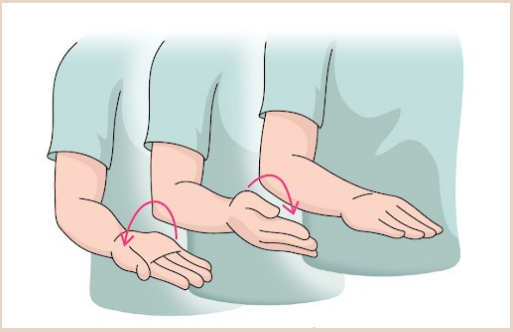
Tenderness, swelling, and heat may be noted with shoulder strains, sprains, arthritis, bursitis, and degenerative joint disease (DJD).

Painful and limited abduction accompanied by muscle weakness and atrophy is seen with a rotator cuff tear. Client has sharp catches of pain when bringing hands overhead with rotator cuff tendinitis. Chronic pain and severe limitation of all shoulder motions are seen with calcified tendinitis.

Inspect, palpate, & ROM: Elbows



Flexion & Extension



Supination & Pronation

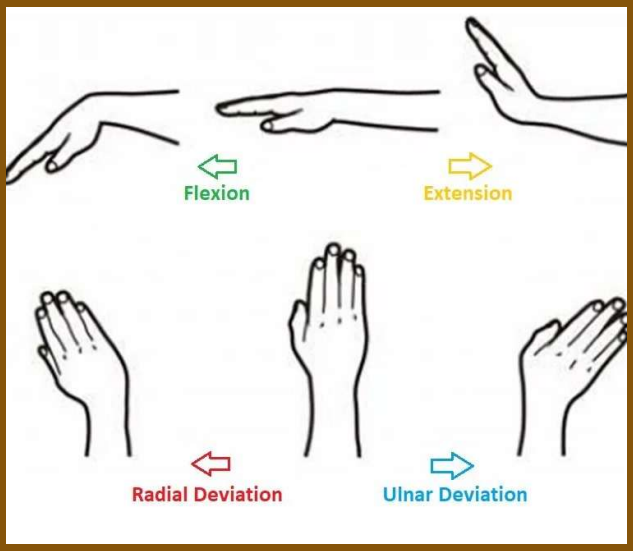
Movements of the elbow: Flexion/extension, supination/pronation

Inspect elbows in both flexed and extended positions.

Elbows are symmetric, without structural abnormalities, redness, or swelling. Nontender; without nodules.

Redness, heat, and swelling may be seen with bursitis of the olecranon process due to trauma or arthritis. Firm, nontender, subcutaneous nodules may be palpated in rheumatoid arthritis or rheumatic fever. Tenderness or pain over the epicondyles may be palpated in epicondylitis (tennis elbow) due to repetitive movements of the forearm or wrists. Decreased ROM against resistance is seen with joint or muscle disease or injury.

Inspect, palpate, & ROM: Wrists



Movements of the wrist: Flexion/extension, hyperextension, adduction, radial/ulnar deviation

Inspect wrist size, shape, symmetry, color, and swelling. Then palpate for tenderness and nodules.

Wrists are symmetric, without redness or swelling. They are nontender and free of nodules.

Swelling is seen with rheumatoid arthritis. Tenderness and nodules may be seen with rheumatoid arthritis. A nontender, round, enlarged, swollen, fluid-filled cyst (ganglion) may be noted on the wrists.

Perform squeeze test. Squeeze the client's hand across the knuckle joints. Client should have no pain. Extreme pain may indicate rheumatoid arthritis or psoriatic arthritis of the hand.

Carpal tunnel testing: Phalen's test

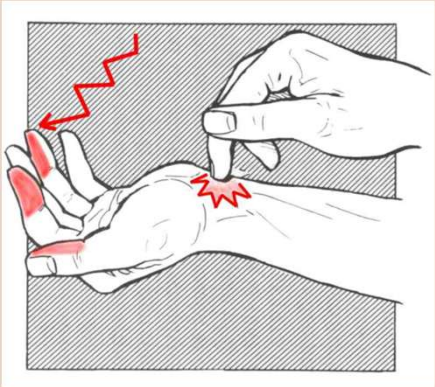


- Client holds hands back-to-back while flexing wrists 90 degrees
- Acute flexion of the wrist for 1 minute produces no symptoms in the normal hand
- Test will reproduce numbness and burning in a client with carpal tunnel syndrome

Phalen test is >85% accurate if the hand position is held for more than 1 minute. However, if the test lasts longer than 1 minute, pain and tingling may occur even in clients without CTS.

Carpal tunnel testing: Tinel's sign

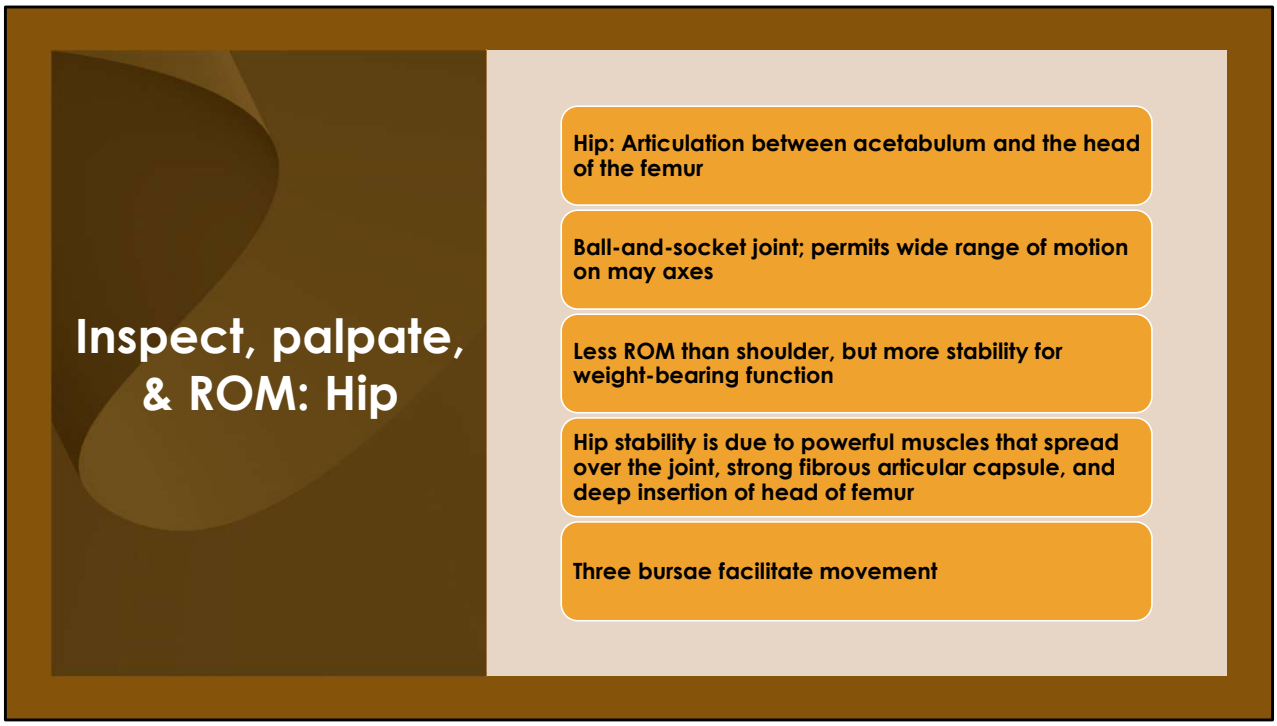
- **Direct percussion of location of median nerve at wrist produces no symptoms in normal hand**
- **Percussion of median nerve produces burning and tingling along its distribution, which is a positive Tinel's sign for carpal tunnel syndrome**



Use your finger to percuss lightly over the median nerve (located on the inner aspect of the wrist).

Median nerve entrapped in the carpal tunnel results in pain, numbness, and impaired function of the hand and fingers.

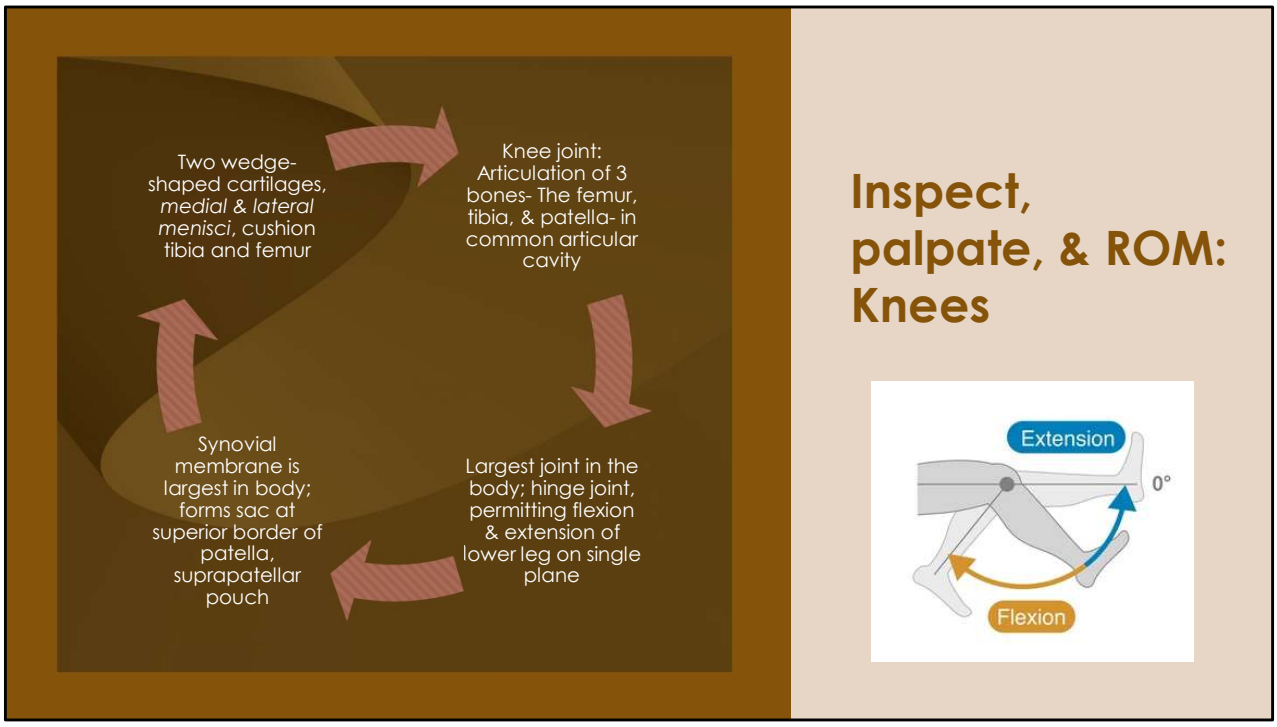
Median nerve distribution: thumb, index finger, lateral ½ of middle finger



Movements of the hip: Flexion with knee flexed & knee extended, extension/hyperextension, circumduction, internal/external rotation, abduction/adduction

With the client standing, inspect symmetry and shape of the hips. Observe for convex thoracic curve and concave lumbar curve. Palpate for stability, tenderness, and crepitus. Hips are stable, nontender, and without crepitus.

Instability, inability to stand, and a deformed hip area are indicative of a fractured hip. Tenderness, edema, decreased ROM, and crepitus are seen in hip inflammation and DJD.



Movements of the knee: flexion/extension

Inspect for size, shape, symmetry, swelling, structural abnormalities, and alignment. Observe for quadriceps muscle atrophy. Palpate for tenderness, warmth, consistency, and nodules.

Knees symmetric, hollows present on both sides of the patella, no swelling or structural abnormalities. Lower leg is in alignment with the upper leg. Nontender and cool. Muscles firm. No nodules.

Knees turn in with knock knees (genu valgum) and turn out with bowed legs (genu varum). Swelling above or next to the patella may indicate fluid in the knee joint or thickening of the synovial membrane. Tenderness and warmth with a boggy consistency may be symptoms of synovitis.

Osteoarthritis is characterized by a decreased ROM with synovial thickening and crepitation. Pain or clicking is indicative of a torn meniscus of the knee.

Inspect, palpate, & ROM: Feet & ankles





Movements of ankles: plantar flexion/dorsiflexion; movements of the feet: Inversion/eversion

Palpate the toes from the distal end proximally, noting tenderness, swelling, bony prominences, nodules, or crepitus of each interphalangeal joint.

No pain, heat, swelling, or nodules are noted.

Ankles are the most common sites of sprains, which occur with stretched or torn ligaments (tough bands of fibrous tissue connecting bones in a joint).
Tender, painful, reddened, hot, and swollen metatarsophalangeal joint of the great toe is seen in gouty arthritis. Nodules of the posterior ankle may be palpated with rheumatoid arthritis.

A laterally deviated great toe with possible overlapping of the second toe and possible formation of an enlarged, painful, inflamed bursa (bunion) on the medial side is seen with hallux valgus.

Tenderness of the calcaneus of the bottom of the foot may indicate plantar fasciitis.

Plantar fasciitis is the most common cause of heel pain, which occurs when the strong supportive band of tissue in the arch of the foot becomes irritated and inflamed.



Older adult considerations

- Greater risk for fractures
- Decreased flexibility
- Impaired sense of position
- Kyphosis is common
- May have bow-legged appearance

Bones lose their density with age, putting the older client at risk for bone fractures, especially of the wrists, hips, and vertebrae. Older clients who have osteomalacia or osteoporosis are at an even greater risk for fractures.

Decreased flexibility may make some positions for exam difficult

Impaired position sense places them at a greater risk for falls

Kyphosis is common with a backward head tilt to compensate

Bow-legged appearance due to osteoarthritis wearing down the cartilage of the medial portion of the knee joint

Older adult considerations: functional assessment

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For those with advanced aging changes, arthritic changes, or musculoskeletal disability, perform functional assessment for ADLs

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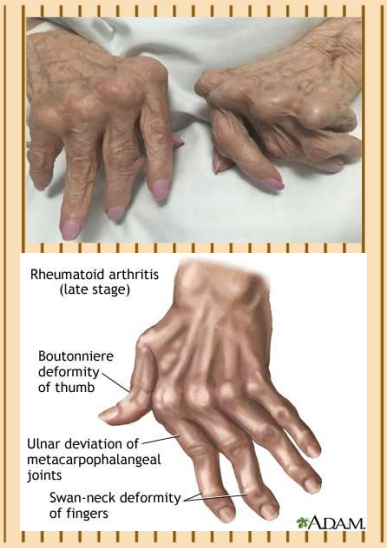
Apply ROM and muscle strength assessments to accomplishment of specific activities

03

Goal is to determine adequate and safe performance of functions essential for independent home life

Abnormalities affecting multiple joints

- Inflammatory conditions
  - Rheumatoid arthritis
  - Ankylosing spondylitis
- Degenerative conditions
  - Osteoarthritis (Degenerative joint disease)
  - Osteoporosis



Rheumatoid arthritis

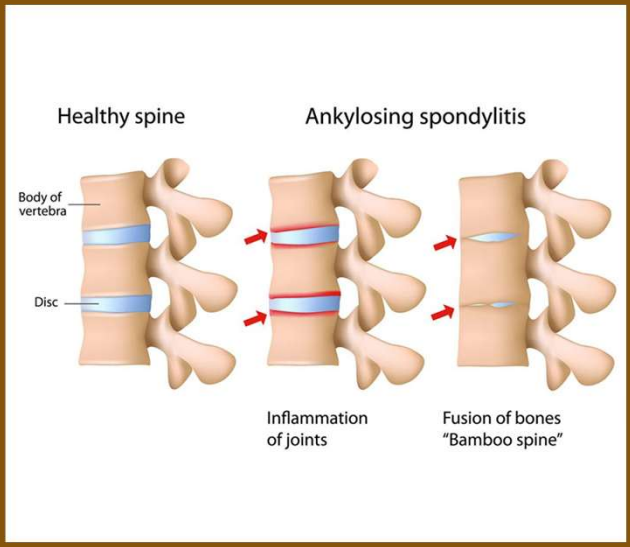
- Chronic autoimmune disease
- Inflammation of joints, synovial tissues, & swelling
- Leads to destruction of cartilage and bone
- Appears as a deformity; finger deviation towards the ulnar side
- May present with fatigue, weakness, anorexia, weight loss, fever, lymphadenopathy

RA: Chronic swelling and thickening of the metacarpophalangeal and proximal interphalangeal joints, limited range of motion and finger deviation toward the ulnar side are seen in chronic RA.
Rheumatoid arthritis pain and symptoms are varied; client may feel burning or throbbing on both sides of the body, which worsen after sitting for long periods, may have inconsistent pattern of worse and less pain, and may experience a feeling of heat and soreness in joints. The client may also experience weak muscles, feeling tired or depressed, weight loss, decreased appetite, slightly elevated temperature, swollen glands, or significant stiffness in the morning that persists at least 1 hour.

Clinical considerations: no heavy resistance training, walking program, pain management education (body mechanics and symptom management strategies)

Ankylosing spondylitis

- Chronic inflammatory arthritis
- Fusion of inflamed vertebrae
- Results in kyphosis
- Affects those assigned male at birth more



An inflammatory arthritis of the spine, hips, & shoulders where inflammation and tendency for fibrosis and calcification at sides of muscle insertion.
Your client with ankylosing Spondylitis will have kyphosis and extension of the neck
Eyes: uveitis(inflammation in the eyes), enthesitis-inflammation and tendency for fibrosis and calcification at sites of muscle insertion.
Extreme loss of mobility of the spine.

Osteoarthritis



- Most common type of arthritis
- Localized
- Progressive, destroying articular cartilages of the joint surfaces
- Incidence increases with age
- Increased stiffness, swelling, & pain with motion
- Over time limitation of movement

Osteoarthritis pain is worst in the mornings and improves with movement.
"The motion is the lotion"

Risk factors:
Age (>50)
Joint injury
Overuse
Obesity
Musculoskeletal abnormalities (Malalignment of bone or joint structures)
Weak muscles
Genetics
Sex assigned at birth (Female more at risk)
Environmental factors (Occupation, level of physical activity, diet, bone density)

Health promotion and disease prevention: Osteoporosis



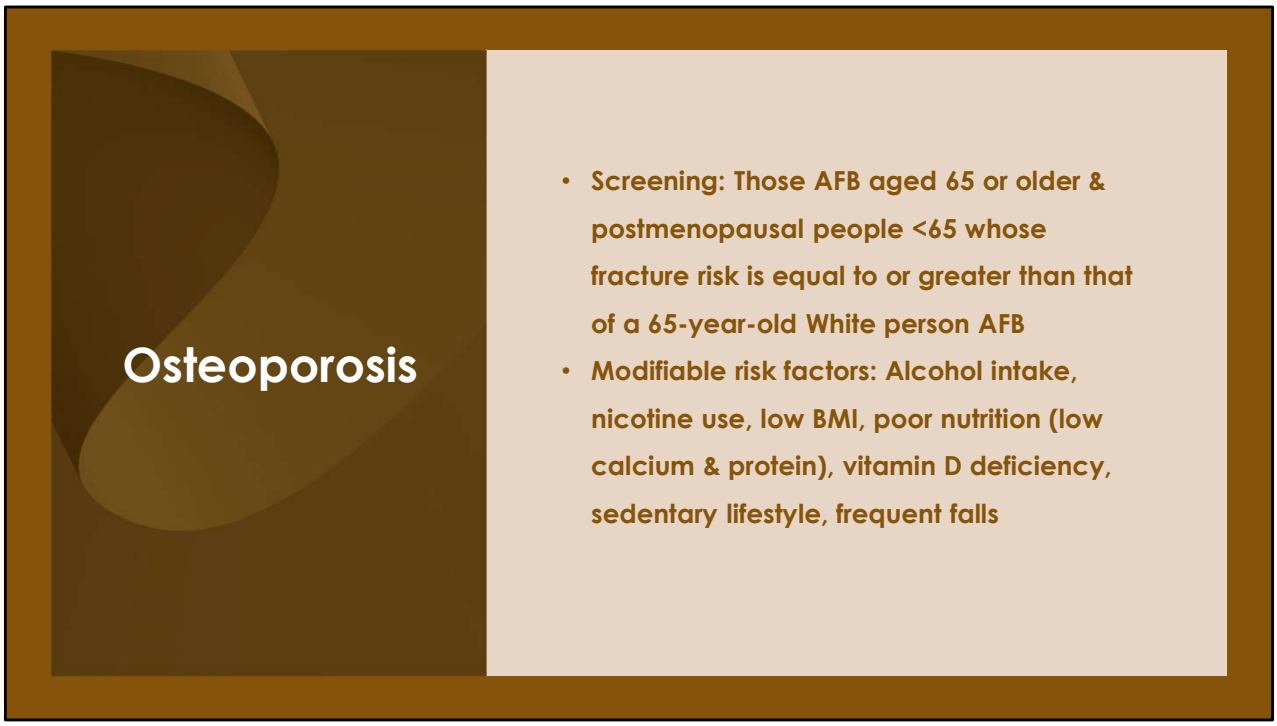
- Loss of skeletal bone mass
- Results in low bone mineral
- Increased risk for fractures: spine, wrist, hip
- Affects 200 million people AFB worldwide
- Causes 8.9 million fractures annually
- Primarily White, post menopausal people AFB

Healthy People 2030 goal:

- Prevent fractures and disabilities related to osteoporosis

Osteoporosis is a disease in which bones demineralize and become porous and fragile, making them susceptible to fractures. The bone loss occurs silently and progressively, and often no symptoms are noted until the first fracture occurs.

Osteoporosis is most common in people assigned female at birth who are non-Hispanic White or Asian, and although Black and Hispanic people assigned female at birth are at lower risk, they are still at significant risk.



Fixed Risk Factors for Osteoporosis and for Fractures:
Age (65 or older)
Assigned female at birth
Family history of osteoporosis or osteoporotic fracture
Previous fracture, including vertebral fracture
Ethnicity (White individuals and Asian individuals are at higher risk)
Menopause/hysterectomy
Long-term use of glucocorticoid therapy such as prednisone
Medical conditions (such as celiac disease, Crohn disease) that inhibit absorption of nutrients
Tendency to fall
Early menopause (before age 45)
Rheumatoid arthritis
Hyperthyroidism
Low bone mineral density



Osteoporosis

Client education:

- Ensure nutritious diet with adequate calcium intake
- Avoid undernutrition
- Maintain adequate supply of vitamin D
- Participate in regular weight-bearing activity
- Avoid heavy drinking
- Avoid active and secondhand smoking

Falls in the elderly





Falls in the elderly

One in four Americans aged 65+ fall each year

Every 11 seconds, an older adult is treated in the ED for a fall; Every 19 minutes, an older adult dies from a fall

Falls are the leading cause of fatal injury and the most common cause of nonfatal trauma-related hospital admissions among older adults

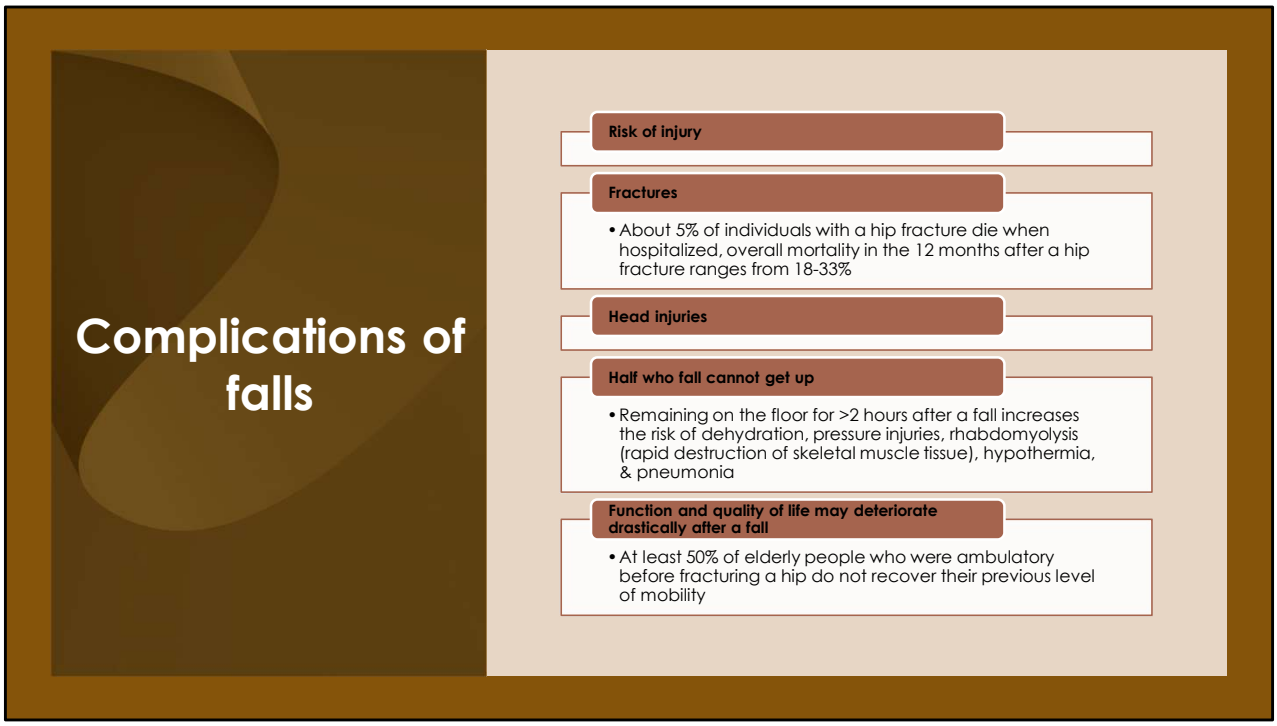
Falling once doubles the chances of them falling again

Etiology of falls



1. **Age-related decline in function, disorders, and adverse drug effects**
2. **Environmental hazards**
   Clutter, loose rugs, cords, small animals
3. **Situational factors**
   Rushing to the bathroom or to answer the door

Adverse drug effects really can contribute to an elderly fall. Just something to think about.. HTN is a very common disease in the older clients…what type of medications do we give them? What is the purpose of those antihypertensives? So what adverse effect do you think a client could experience at home?
Can anyone tell me some environmental hazards?
Situational factors: Older clients experience urgency and frequency, so they rush to the bathroom, reaching for furniture to grab..

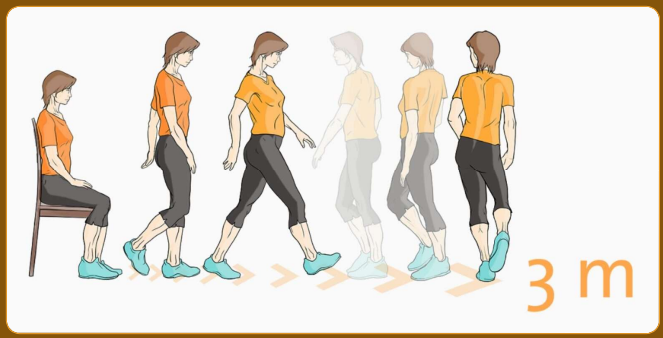


Once a client has fallen, they put themselves at risk for falling again.
Once they have fallen the older client has a significant fear of falling again.. Especially if they had a previous injury due to the fall.

Many older adults are reluctant to report a fall due to fear of losing independence or being institutionalized.

The timed up and go test (TUG)

- Tests mobility
  - Observation of client's gait
- Timed test
- Person rises from chair
- Walks 10 feet, turns
- Walks back to chair
- Sits down

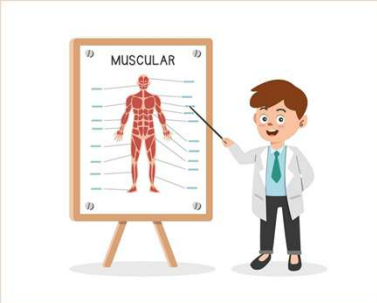


Directions: Patients wear their regular footwear and can use a walking aid, if needed. An older adult who takes ≥12 seconds to complete the TUG is at risk for falling.

OBSERVATIONS:
Observe the patient's postural stability, gait, stride length, and sway.
Slow tentative pace
Loss of balance
Short strides
Little to no arm swing
Steadying self on walls
Shuffling
Not using assistive device properly
These changes may signify neurologic problems that require further intervention.

https://www.youtube.com/watch?v=tNay64Mab78

Review question



A nurse is caring for a client who reports pain with internal rotation of her right shoulder. This discomfort can affect the client's ability to perform which of the following activities?

A. Mopping her floors

B. Brushing the back of her hair

C. Fastening her bra behind her back

D. Reaching into a cabinet above the sink

A client who is fastening a bra from behind requires internal rotation of the shoulder

A. Is extension and flexion of the shoulder

B. The client who is brushing the back of the hair requires external rotation of the shoulder

D. The client who is reaching for something up high requires external rotation of the shoulder.

Review question

When reviewing home safety with a client and their family, which modifications can reduce the client's risk for falls? Select all that apply.

A. Remove all types of rugs

B. Place grab bars near the toilet and shower

C. Ensure that lighting is adequate inside and outside the home

D. Add tennis balls to the bottom of the walker if they use one

E. Place furniture close enough for the client to hold to ambulate around the house

All